

Predicting cross-national sex differences in large-scale assessments of students' reading literacy, mathematics, and science achievement: Evidence from PIRLS and TIMSS

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ABSTRACT

Causes of sex differences in educational achievement have been controversially discussed in the extant literature. It has been speculated that differing prosperity and equality of opportunities may be linked to these differences, but conclusive empirical evidence for such effects is unavailable. Here, we present evidence for sex differences in international large-scale assessments of reading literacy, mathematics, and science across 16 cohorts from 1995 to 2019. Our analyses of PIRLS and TIMSS reading literacy, mathematics, and science achievement data ($N = 3,999,062$; 90 countries) showed consistent advantages for girls in reading literacy (d range: -0.02 to 0.66). For mathematics and science this pattern was less unambiguous, yielding non-trivial effects in both directions (d ranges: -0.44 to 0.36 and -0.50 to 0.46 , respectively). Sex differences in all three domains were more pronounced in more egalitarian countries (β range 0.16 to 0.20). Higher national prosperity and educational investment predicted larger sex differences favoring fourth grade boys in mathematics and science (β range: 0.07 to 0.39) and became less meaningful with increasing student ages (β range for eighth graders: 0.17 to 0.21). In all, our findings suggest that influences of economic macro-indicators on sex differences in educational achievement are differentiated according to subject, indicating larger sex differences in mathematics and science in more egalitarian and prosperous countries.

1. Introduction

Sex differences have been observed in a majority of researched domains, spanning a range of effect sizes from small (e.g., object location memory, negative emotions) to medium (e.g., anxiety disorders, impulsivity) and even large (e.g., social interests, sexual violence) effects, depending on the scrutinized trait (Archer, 2019). The accumulated evidence about sex differences in cognition over the past century suggests that they are robust in specific domains, although they do not seem to map onto psychometric g (Reynolds, Hajovsky, & Caemmerer, 2022; for some contrasting evidence see Johnson, Carothers, & Deary, 2008). This corresponds to the ideas of the Gender Similarities Hypothesis (GSH; Hyde, 2014), which states that there are no sex differences in general cognitive ability, but that there are consistent differences in specific cognitive domains.

Sex differences in academic achievement and related cognitive abilities (e.g., spatial ability) have been studied for decades. There is a large body of evidence suggesting better performance of girls than those

of boys in reading literacy (Stoet & Geary, 2013). For example, evaluations of the Programme of International Student Assessment (PISA) of 2000, 2003, and 2006 showed that girls typically obtained higher average scores in reading ability than boys ($d = 0.42$; Lynn & Mikk, 2009). This is supported by meta-analytical findings of sex differences in educational achievement which showed advantages of girls in reading grades ($d = 0.37$; Voyer & Voyer, 2014). Recent studies suggest that girls' advantages are most prominent in terms of reading fluency, although these advantages appear to generalize to all reading domains (Reynolds et al., 2022).

For mathematics and science, opposite patterns were typically observed. Past research showed higher achievement scores of boys compared to girls in both mathematics and science (for an overview, see Reilly, Neumann, & Andrews, 2015). Spatial abilities appear to contribute to boys' advantages in some areas of mathematics (Halpern et al., 2007). Even so, there may be no sex differences in mathematics domains that do not have a strong spatial component or can be solved with standard solution strategies that are being taught in school

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(Gallagher & De Lisi, 1994). This may lead to discrepancies in achievement sex differences effect strengths and signs depending on the respective investigated mathematics domain. Indeed, between-country analyses revealed a considerable variability of effect sizes ($d = -0.42$ to 0.40 ; Else-Quest, Hyde, & Linn, 2010), indicating that sex differences in mathematics cannot be assumed to behave in an identical fashion across different countries. Thus, whether boys outperform girls in mathematics and science could represent consequences of differing sociocultural and economic conditions between nations, comprising a country's societal norms, gender equality, liberality, and prosperity.

Previous evidence supports this idea. Specifically, results based on the 2003 PISA cohort indicate, that sex differences in mathematics are substantially lower in more gender-egalitarian countries, but girls still outperform boys in reading literacy (Guiso, Monte, Sapienza, & Zingales, 2008). However, these findings have not been replicated in more extensive datasets (PISA and TIMSS data from 2000 to 2015), suggesting that more gender-egalitarian national macro-indicator values are associated with improved performance of boys relative to girls in reading literacy, mathematics, as well as science (Eriksson, Björnstjerna, & Vartanova, 2020). In the light of such inconsistent observations, it has been argued that effects of national macro-indicators of gender equality on achievement sex differences may be driven by specific characteristics of different achievement assessments and are mainly due to outliers (Stoet & Geary, 2015).

In terms of general socioeconomic and cultural equality, cross-sectional findings of PISA indicated, that sex differences in the high-achievement PISA segment (i.e., students at levels 5 or 6 from the six PISA proficiency levels) were smaller in more compared to less egalitarian countries in reading literacy, mathematics, and science (Breda, Jouini, & Napp, 2018). In turn, intra-individual ability tilt (tilt = mathematics scores – verbal scores) appeared to have been unrelated to nationality (Wai, Hodges, & Makel, 2018).

Furthermore, higher achievement scores of boys in mathematics and science have been found to correlate positively with economic prosperity (i.e., gross domestic product per capita), whilst girls' advantages in reading literacy were unrelated to economic prosperity (Reilly, 2012). Similar results have been observed in formal achievement assessments in Italy where sex differences in mathematics performance, but not in reading literacy, have been found to be linked to macro-indicators of national and regional prosperity (Giofre, Cornoldi, Martini, & Toffalini, 2020).

These results suggest that sex differences in mathematics and science may be expected to be larger in more prosperous countries, whilst reading literacy should remain unaffected. It has been speculated that these prosperity-related sex differences in mathematics performance could be a consequence of larger socioeconomic inequality in less prosperous regions (Giofre et al., 2020). Furthermore, it can be argued that higher economic prosperity and material security may contribute to stronger gender-essentialist ideologies (i.e., the belief that particular traits are inherently masculine or feminine), as well as stereotypes through the pursuit of educational and occupational goals as reflection of self-referential norms (Charles & Bradley, 2009; Miller & Halpern, 2014). However, other researchers have suggested that achievement-based sex differences might be more reasonably explained by nature-rather than nurture-dependent influences (e.g., genetic bases in general: Schmitt et al., 2017; increased opportunity to express inherent preferences; Stoet & Geary, 2018).

Conceivably, further socioeconomic factors that relate to national prosperity and economic inequality in general and educational spending in particular may play a further role in between-countries variations of achievement-based sex differences that has yet to be clarified. To contribute towards identifying predictors of sex differences, we use data from PIRLS (Progress in International Reading Literacy Study) and TIMSS assessments (Trends in International Mathematics and Science Study), which include almost 4,000,000 participants from more than 90 countries. To disentangle cohort and age effects, we analyze data from

several waves and grades.

To date, most available investigations on school achievement-based sex differences have relied on data from the PISA assessment. Presently, we examine sex differences on two other formal international large scale assessments of school achievement which represent dedicated evaluations of students reading literacy (PIRLS) and mathematics/science achievement (TIMSS). In this vein, we use several indicators of national prosperity and economic (in-)equality to predict sex differences in different achievement domains and investigate possible (i) associations with indicators of equal education opportunities (presently assessed by the Gender Gap Index and Gender Parity Index) as well as (ii) links with educational spending.

We expected to observe higher performance of girls than of boys in reading literacy but larger scores of boys than of girls in mathematics and science. Moreover, we expected larger sex differences in more prosperous and egalitarian countries as well as in those that invest more in education for mathematics and science but not for reading literacy.

2. Methods

The present study was preregistered prior to accessing any data. The preregistration protocol, supplementary materials, presently used data alongside the respective analysis code, as well as any deviations from the preregistration are available in an online repository (<https://osf.io/thrjn> and <https://osf.io/3qjrd/>).

Student achievement data were retrieved from the data repository of the IEA (International Association for the Evaluation of Educational Achievement; <https://www.iea.nl/data-tools/repository>). Responses on both PIRLS and TIMSS assessments included multiple choice as well as open response formats. The grading of open responses was carried out by experienced staff and was subject to rigorous control processes. All cohort scores are referenced to a mean value of 500 ($SD = 100$) representing the average performance of the first cohort (i.e., PIRLS: 2001; TIMSS: 1995) of participants from all countries in grades 4 and 8 that had participated in this initial cohort assessment. All subsequent scores are referenced to this initial standardization and cross-temporal measurement invariance of assessments between different cohorts is evaluated, which enables comparisons of cross-national and cross-temporal performance.

Data for the national socioeconomic indicators used in our study were retrieved from the World Bank (<https://data.worldbank.org>), excepting data for Chinese Taipei, which was retrieved from <https://www.statista.com>. Year- and country-specific macro indicator values were used as predictors in our regression models.

In all, 3,999,062 participants from 90 countries were included in the present analyses. Country-level participant numbers according to cohort are provided in the online Supplementary Materials (see Tables S1-S4, <https://osf.io/73n8k>).

2.1. PIRLS (Progress in International Reading Literacy Study)

To investigate sex differences concerning reading literacy achievement, we used data from four cohorts of the PIRLS assessments from 2001 to 2016 (data were assessed in five-year intervals; i.e., 2001, 2006, 2011, 2016). In the PIRLS assessments, reading literacy is assessed in fourth grade students (i.e., 10-year-olds). In these assessments, students must first read texts and subsequently respond to questions that are related to these texts. Students are required to respond to multiple-choice items (scored as either fully correct or incorrect) as well as open-response questions (scored as either acceptable vs. unacceptable responses, or not administered) in a pen and paper format. In this vein, three main aspects of reading literacy are measured from which an overall score is derived, namely (i) reading intentions, (ii) comprehension processes, and (iii) reading behavior and attitudes towards reading.

To assess the first aspect (i.e., reading intentions), both reading to gain information (informational texts) and reading to gain literacy

experience (literary texts) are used. The second aspect (i.e., comprehension processes) refers to the cognitive skills students need to apply in order to answer questions about the texts they have read (i.e., recognizing and reproducing information, drawing inferences, interpreting information, and evaluating content and language). Finally, students self-report their reading attitudes and behavior. Depending on the administration year, PIRLS reading literacy assessments comprised 98 to 135 items, averaging a testing time of two and a half hours (assessment procedures are detailed at <https://www.iea.nl/studies/iea/pirls>).

2.2. TIMSS (Trends in International Mathematics and Science Study)

We used data from five and seven cohorts (grade 4: 10-year-olds; 2003 to 2019; grade 8: 14-year-olds; 1995 to 2019; data were assessed in four-year intervals) of the TIMSS assessments, respectively, to investigate sex differences in mathematics and science. In TIMSS, three superordinate cognitive domains (knowing, applying, and reasoning) are assessed in both mathematics and science achievements of fourth- and eighth-graders. Students are required to respond to multiple choice items (scored as either fully correct or incorrect) as well as open-response questions (scored as either fully correct, correct, or incorrect). These items are grouped in a series of blocks for different content areas (i.e., 10 to 14 and 12 to 18 items per area for fourth- and eighth-graders, respectively).

Specifically, overall scores for fourth graders performance in mathematics are determined by three content domains, namely (i) number, (ii) geometric shapes and measurement, as well as (iii) data display, whilst overall scores for science achievement are derived from (i) life science, (ii) physical science, as well as (iii) earth science. For eighth graders, mathematics overall scores are obtained from four content domains, which are (i) number, (ii) algebra, (iii) geometry, as well as (iv) data and chance, whilst eighth graders science achievement overall scores are based on (i) biology, (ii) chemistry, (iii) physics, as well as (iv) earth science. Testing time averages two and a half hours (assessment procedures are detailed at <https://www.iea.nl/studies/iea/timss>).

2.3. Macro-level predictors

For each included country, indicators in terms of prosperity, economic equality, and educational spending (i.e., GDP per capita - GDP; Gender Gap Index - GGI: a quantitative measure to assess (in-)equality between girls and boys in terms of access to resources in four domains, namely health, education, economy, and politics with 0 indicating perfect inequality and 1 perfect equality; Gini Index: higher values indicate more inequality; Gender Parity Index for primary and secondary school enrollment - GPI: representing the school enrollment ratio of girls and boys on both primary and secondary school level: <1 = more boys, >1 = more girls; government expenditure on education expressed as a percentage of GDP - GET; government expenditure per primary/secondary student expressed as a percentage of GDP - GEP/GES) were retrieved for the respective assessment years. If no index was accessible for the exact year in which student achievement was obtained, the most proximal annual value was used (see <https://osf.io/3qjrd/> for a detailed documentation).

2.4. Data analyses

In our main analyses, we ran all calculations including only data from those countries, that had participated in each year of assessment (see Table 1 for an overview of included years and countries).

To examine the stability of our results, we reran all analyses by including all countries, regardless of the number of cohorts that a respective country provided data for (see Tables S5-S21; <https://osf.io/rnbme>).

Table 1

Data assessment years and included countries for main analyses.

	PIRLS	TIMSS 4	TIMSS 8
Year			
1995			•
1999			•
2001	•		
2003		•	•
2006	•		
2007		•	•
2011	•	•	•
2015		•	•
2016	•		
2019		•	•
Country			
Armenia		•	
Australia		•	•
Bulgaria	•		
Canada	•	•	•
Chinese Taipei		•	
France	•		
Germany	•		
Hong Kong SAR	•	•	•
Hungary	•	•	•
Islamic Republic of Iran	•	•	•
Israel	•		•
Italy	•	•	•
Japan		•	•
Lithuania	•	•	•
Morocco	•	•	
Netherlands	•	•	
New Zealand	•	•	
Norway	•	•	
Republic of Korea			•
Russian Federation	•	•	•
Singapore	•	•	•
Slovak Republic	•		
Slovenia	•		
Sweden	•		
United Kingdom	•	•	•
United States	•	•	•

Note. "•" indicates that the respective year or country was included in the main analyses.

2.4.1. Standardized mean sex differences

We calculated sex differences of overall scores for reading literacy, mathematics, and science to compute standardized mean differences (Cohen *ds*) between the performance of boys and girls (positive values indicate higher scores of girls, negative values indicate higher scores of boys) within each country. The outcome measures that are available for PIRLS and TIMSS assessments for each (sub-)scale are calculated according to different (but largely similar) procedures that correct for potential sampling artefacts (e.g., variance restriction; data nesting of students within the same school), yielding five so-called plausible values that are available for each surveyed year (Fishbein, Foy, & Yin, 2021). We calculated standardized mean sex differences for each plausible value and subsequently averaged them for our reporting. Numerical values for standardized sex differences of individual plausible values as well as cohort-specific results are provided in Tables S22-S36 at <https://osf.io/5uqkj> (results were virtually identical to overall estimates).

2.4.2. Associations of national indicators with sex differences

To examine the influence of national indicators on sex differences, we conducted a series of single regression analyses (this approach was adopted to alleviate effects of expectable multicollinearity of our predictors, following the rationale of Agresti & Finlay, 2009, p.457; this approach was preregistered at <https://osf.io/thrjn>). Accordingly, GDP per capita, GGI, Gini Index, GPI, GET, and GEP/GES were used as predictors for sex differences (i.e., Cohen *ds* of the respective plausible

values) in individual single regressions. To avert the possibility of underestimating standard errors, we carried out regression analyses on each plausible value and combined the outcomes based on Rubin (1987) rules. Country-specific regressions yielded virtual identical results and are provided in Table S37, available at <https://osf.io/cn6gp>.

In order to capture the common variance of the national socioeconomic indicators, we used the approach of Revelle and Wilt (2013) to conduct a latent variable exploratory factor analysis (EFA) using GDP per capita, the (reverse-coded) Gini Index, GPI, GGI, and GEP/GES. In this vein, we multiplied the z-standardized scores of the socioeconomic indicators of an exploratory single-factor principal component analysis by their respective loadings. Subsequently, we calculated a weighted average of the five resulting scores which can be interpreted as a composite of the national socioeconomic indicators. The direction and magnitude of the effects were largely identical to the findings in our main analysis (the intercorrelation matrix of the predictors and the results of the regression analyses for reading literacy, mathematics, and science are available at <https://osf.io/3qjrd/files/osfstorage>).

In our results, we focus on the interpretation of effect sizes according to well-established effect size thresholds (i.e., Cohen d s: 0.2, 0.5, and 0.8; Pearson's r : 0.1, 0.3, and 0.5; as lower thresholds of small, moderate, and large effects, respectively; Cohen, 1988) instead of outcomes from conventional null-hypothesis significance testing (Cumming & Calin-Jageman, 2016). Within our regression analyses, the meaningfulness of the strength of standardized regression coefficients can be interpreted according to the same criteria as correlations coefficients (Pearson's r). All statistical analyses were conducted in R version 4.0.2 (R Core Team, 2022), and RStudio 2022.07.2 + 576 (RStudio Team, 2020).

3. Results

3.1. Standardized mean sex differences

We provide in our online Supplementary Materials sex differences for individual countries within cohorts, grades, and domains (Figs. S1-S28 <https://osf.io/m56gq>) as well as country-, grade-, and domain-specific global heat maps of average sex differences (Figs. S29-S33 at <https://osf.io/fydzr>).

Standardized mean sex differences (Cohen d) of all countries that had participated in each year of assessment are illustrated in Fig. 1. We observed a consistent pattern of advantages for girls compared to boys in terms of reading literacy across all cohorts (mean d range: 0.19 to 0.23). In mathematics as well as science, boys showed on average consistently higher performances than girls in all cohorts, although effect sizes were small-to-trivial in strength (mean d range: -0.23 to < 0.01). However, country- and cohort-specific analyses revealed differences in performance with meaningful advantages for both sexes, although no clear pattern emerged favoring either sex (i.e., whether boys or girls demonstrated better performance varied across different countries and cohorts; see Supplementary Figs. S1 to S33 at <https://osf.io/m56gq> and <https://osf.io/fydzr>).

3.2. Influences of national macro-indicators on sex differences

Regression outcomes for reading literacy, mathematics, and science in both fourth and eighth grades are provided in Fig. 2. Detailed numerical results can be obtained from Table S38 at <https://osf.io/k7y8n>.

Regression analyses for reading literacy revealed negative small effects of GDP per capita ($\beta = -0.21$) and the Gini Index ($\beta = -0.19$). These findings indicate that higher national prosperity and lower economic equality are negatively related to girls' advantages in reading

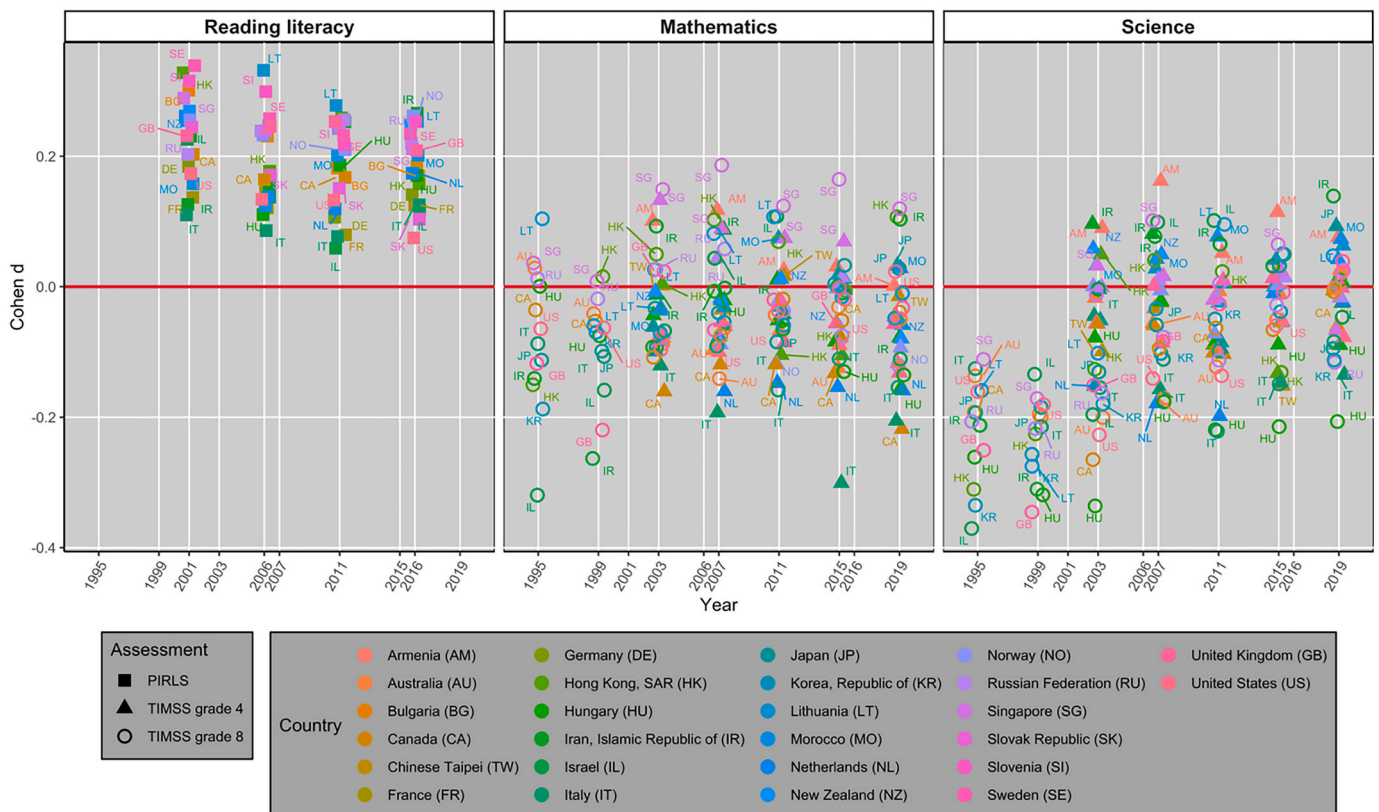


Fig. 1. Standardized mean sex differences for reading literacy, mathematics, and science in fourth and eighth graders within assessment years according to country. Note. Positive values indicate higher scores of girls, negative values indicate higher scores of boys.

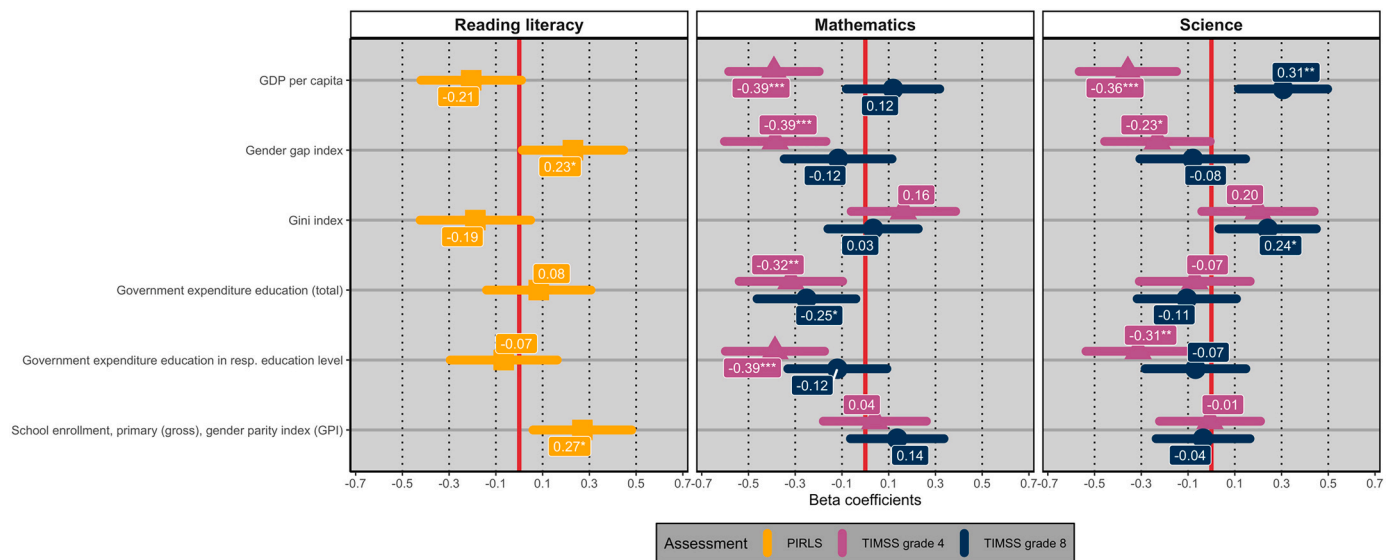


Fig. 2. Effects of national indicators on standardized sex differences in reading literacy, mathematics, and science for fourth and eighth graders.

Note. Values represent standardized single regression weights. *) $p < .05$; **) $p < .01$; ***) $p < .001$. For reading literacy, negative effects indicate smaller, whilst positive effects indicate larger advantages for girls compared to boys (except for the Gini index because it is coded inversely). For mathematics and science, negative effects indicate larger, whilst positive effects indicate smaller advantages for boys compared to girls (except for the Gini index because it is coded inversely).

literacy. The Gender Gap Index (GGI) as well as the Gender Parity Index for primary and secondary school enrollment (GPI) showed a positive small to medium-sized effect ($\beta = 0.23$ and $\beta = 0.27$, respectively). This indicates that girls' advantages in reading literacy are higher in more gender equal countries. Government expenditure on total education (GET) as well as per primary student was unrelated to performance sex differences in reading literacy ($\beta = 0.08$ and $\beta = -0.07$, respectively; see Fig. 2), suggesting no meaningful effects of educational government expenditure on sex differences in reading literacy.

For mathematics in grade four, we observed a consistent pattern of meaningful negative effects of the GDP per capita ($\beta = -0.39$), the GEP ($\beta = -0.39$), the GET ($\beta = -0.32$) as well as the GGI ($\beta = -0.39$). This indicates that higher national prosperity, larger educational investment as well as more gender equality in health, education, economy, and politics, were associated with larger performance advantages for boys. A positive, small effect was observed in terms of the Gini Index ($\beta = 0.16$), suggesting larger mathematics performance advantages of boys over girls in more egalitarian countries. The GPI did not meaningfully predict sex differences in fourth graders' mathematics scores ($\beta = 0.04$).

For mathematics in grade eight, we observed a small to medium-sized, negative effect of GET ($\beta = -0.25$), indicating that larger educational investment was linked to larger advantages in mathematics performance of boys over girls. This is consistent with our observation of a small effect of governmental expenditure per secondary student (GES; $\beta = -0.12$). GDP per capita ($\beta = 0.12$) as well as GPI showed a small positive effect ($\beta = 0.14$), indicating smaller advantages of boys over girls in more prosperous countries with higher gender parity in terms of school enrollment. For the GGI, analyses revealed a small negative effect ($\beta = -0.12$), indicating better performances of boys over girls in countries with larger gender equality. The Gini Index ($\beta = -0.03$) showed no meaningful effect.

Similar to mathematics, we observed medium-sized, negative effects of GDP per capita ($\beta = -0.36$) and GEP ($\beta = -0.31$) for science in grade four. This indicates positive effects of national prosperity and primary school educational investment on the performance of boys. Furthermore, we found a small, positive effect of the Gini Index ($\beta = 0.20$), indicating better performances of boys in science in more egalitarian countries. Again, similar to mathematics, the GGI showed a meaningful effect ($\beta = -0.23$), suggesting a small positive influence of gender equality on the performance of boys in science. We did not find

meaningful effects of GEP as well as GPI (β range: -0.07 to -0.01) for science in grade four.

For science in grade eight, GDP per capita and the Gini Index showed positive small to medium-sized effects ($\beta = 0.31$ and $\beta = 0.24$, respectively), suggesting smaller performance sex differences in science and in more prosperous, but less egalitarian nations. Analyses revealed a negative small effect of the GET ($\beta = -0.11$) and a trivial negative one of GEP ($\beta = -0.07$), indicating positive influences of the amount of public investment in education on the science performance of boys. The GGI and the GPI were unrelated to sex differences in grade eight science ($\beta = -0.06$ and $\beta = -0.08$, respectively).

To examine potential influences of between-country differences in compulsory schooling policies, we repeated our analyses for only those countries that mandate schooling up to at least grade eight (i.e., age 14). Results were virtually identical to those from the main analysis (<https://osf.io/3qjrd/files/osfstorage>), thus indicating no meaningful effects of schooling mandates on our findings.

Furthermore, we investigated potential effects of students at both ends of the performance distribution. We re-analyzed the bottom and top 25% of worst and best performers in each domain and class (selected by the first plausible value). In general, effects were smaller in data subsets compared to the entire data set, although signs remained identical. However, these numerically weaker effects were expectable and may be reasonably explained by well-known effects of variance restriction (Schmidt & Hunter, 2015; for numerical results, see <https://osf.io/3qjrd/files/osfstorage>).

4. Discussion

Here, we show evidence for cross-temporally persisting differences in educational achievement between girls and boys that appear to be differentiated according to the examined cognitive domain. Importantly, these achievement differences seem to be linked to different macro-indicators of national prosperity, economic and gender (in-) equality, and educational investment. These findings present several points of interest, as we detail below.

Girls showed better performance in reading literacy whilst for mathematics and science, a less unambiguous pattern emerged. Whilst girls outperformed boys in reading across all cohorts in all countries, for mathematics and science, no consistent pattern of sex differences in

either direction emerged. Sex-specific advantages appeared to be differentiated according to the investigated country. Our results indicate that these sex-related differences are linked to country-level socioeconomic factors.

First, girls showed remarkably consistent advantages compared to boys in reading literacy across cohorts in all examined countries, yielding merely a single data point out of 168 with an inconsistent sign which yielded a trivial effect. This overwhelming evidence for girls' reading literacy advantages supports results of prior findings indicating higher girls reading achievement (e.g., [Lynn & Mikk, 2009](#); [Voyer & Voyer, 2014](#)). National prosperity and equality negatively predicted girls' advantages, indicating smaller advantages of girls in more prosperous and less egalitarian countries.

Nonetheless, girls' reading performance was positively influenced by higher equality in terms of (i) the gender gap in health, education, economy, and politics and (ii) relative access to the educational system between boys and girls. However, effects of these socioeconomic indicators were predominantly small and considerably less powerful than their effects on mathematics and science. Furthermore, we did not find any associations of performance differences in reading literacy with governmental expenditure in education (see [Reilly, 2012](#) and [Giofre et al., 2020](#), for similar results).

Second, on average, sex differences in mathematics and science across cohorts were trivial in size. However, sex differences appeared to be substantially differentiated according to the examined countries which supports prior results of cross-national sex differences ([Stoet & Geary, 2013](#)): Whilst in many countries, boys outperformed girls in mathematics, effect sizes in several countries yielded girls' advantages across cohorts.

Our presently observed between-country variations thus support the idea of socioeconomic macro indicators as important drivers of sex differences in mathematics and science. In fourth graders, sex difference variations in signs and strength were meaningfully predicted by national prosperity and societal equality indicators. Boys' advantages seemed to be larger in more prosperous nations as well as more economically and gender equal societies. The amount of public investment in education predicted boys' advantages in both mathematics and science positively, yielding a medium-sized effect.

This is consistent with previous research, indicating that socioeconomic factors may be associated with sex differences in mathematics ([Giofre et al., 2020](#)). Furthermore, individual differences are typically observed to be more pronounced in more egalitarian environments due to reduced environmental variability (e.g., increases in intelligence heritabilities in more egalitarian countries; [Pietschnig, Traub, Siegel, & Pellegrini, 2021](#)).

In contrast to our expectations, prosperity-related macro-indicators showed less meaningful systematic influences for eighth graders in either mathematics or science compared to fourth graders. Conceivably, these results are a consequence of children in more prosperous and egalitarian nations to specialize earlier in their areas of interest, thus mitigating the impact of socioeconomic factors at older ages. We speculated, that the observed associations in eighth grade could be explained by the interaction between a country's prosperity or equality and the enrollment of students in secondary schools. In comparison to fourth graders, there are fewer countries that have compulsory schooling for eighth graders, which leads to a restriction of variance and therefore potentially smaller effects. However, when we reanalyzed our results without such countries, the effects remained virtually unchanged.

Effects for grade eight were smaller in size, in most cases they yield similar implications as those in grade four. However, whilst mathematics achievement advantages of boys appeared to be mostly driven by more public investment in the educational system, boys' science achievement advantages were mostly related to higher economic equality. Interestingly, no consistent effect was found for GDP per capita in the eighth grade. However, this is not unexpected because GDP per capita can be assumed to have smaller effects on scholastic achievement

compared to other prosperity-related measures such as educational spending. Countries with the highest GDPs are not necessarily those that invest most in their educational systems. However, based on our current findings, we cannot be entirely certain if different changes would manifest when school grades with different in-between intervals or initial assessment ages were to be examined.

Our results show that national prosperity and economical as well as gender equality appear to play an important role for achievement differences between boys and girls, particularly when it comes to mathematics in fourth graders. Consistent with previous research, sex differences, especially in mathematics and science, seem to become more pronounced, the wealthier and more egalitarian a country is (e.g., [Giofre et al., 2020](#)). The most consistent influences of any of our investigated predictors were observed in mathematics and science for public investment in education as well as equality for grade four, which showed unequivocally larger sex differences in nations with larger economic and gender equality and higher educational spending. Moreover, reading literacy seemed to be further affected by the relative access of boys and girls to education.

The present findings of national macro-indicator effects on sex differences in reading literacy, mathematics, and science may be attributed to differing possibilities in pursuing one's interests depending on socioeconomic circumstances in a given environment. In other words, individuals may have better opportunities to engage in what they are interested in, when they are living in a rather prosperous and egalitarian society. In this sense, more developed and prosperous countries may allow for a greater expression of individual interests which may in turn exacerbate sex differences, because boys and girls may differ therein ([Giofre et al., 2020](#)).

Sex-dependent interest differences have often been explained according to biopsychosocial explanations ([Miller & Halpern, 2014](#)). In this vein, biological differences (e.g., in brain structure and function) can affect a student's learning, perception, and information processing but also cultural factors such as gender norms and stereotypes (which were argued to be stronger in more prosperous nations; see [Charles & Bradley, 2009](#); however there is no consensus on this and the available empirical evidence is weak; [Howell & Klein, 2018](#)) may influence how boys and girls are encouraged to pursue certain subjects or activities.

Moreover, our findings suggest that national prosperity as well as economic and gender equality are strong facilitators of mathematics and science performance among boys in fourth, but to a considerable smaller extent, in eighth grade.

4.1. Limitations

In the context of the present study, potential influences of country-specific curricula on reading literacy (e.g., [van Hek, Buchmann, & Kraaykamp, 2019](#)) were not investigated. However, we could presently show that country-level macro indicators (e.g., GDP per capita, GGI) can be considered to be meaningful predictors of reading literacy. It needs to be kept in mind when interpreting effects on reading literacy, that PIRLS subscales include a measure on reading attitudes, which is arguably not an objective measure of reading achievement.

Of note, TIMSS-science results in the two initial cohorts (i.e., 1995 and 1999) showed a markedly different sex difference pattern than in later cohorts. This may indicate that revisions of the original assessment instrument may have led to a change in the properties of the observed science (sub-)domains. Interestingly, these sex difference-specific between-countries asymmetries were more substantial for science than for mathematics in the first two assessments, but began to behave in a similar fashion starting from the third assessment.

5. Conclusion

We show here that the malleability of sex differences in educational achievement is differentiated according to domain. Whilst mathematics

and science sex differences appear to be considerably influenced by national prosperity indicators, reading literacy results are less affected. In fact, reading literacy differences in the present study were mainly driven by (economical) gender equality. Whilst girls consistently outperform boys in reading, boys' advantages in mathematics and science achievement differences are rooted in higher country prosperity and more educational spending.

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Declaration of Competing Interest

We have no competing interests to declare.

Data availability

Data will be made available on request.

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